Intra- and Intersource Interactive Representation Learning Network for Remote Sensing Images Classification

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***Abstract*— Recently, remote sensing technology has developed faster and faster, and obtaining high-quality panchromatic (PAN) and multispectral (MS) images has become more accessible. The complementarity between them provides new opportunities in multisource remote sensing image classification. However, solving the problem of the semantic gap between multisource high-level features and, at the same time, utilizing the complementary properties between them to reduce intersource information redundancy is still a challenge. This article constructs an I^3RL-Net for the multisource remote sensing image classification task. Specifically, we design a cross-source interactive enhanced fusion module (CIEF-Module). For multilevel multisource features, by strengthening the dependencies of intrasource features and conducting intersource enhanced fusion, intrasource correlation features are refined, and the problem of the intersource semantic gap can be effectively alleviated. During the cross-source interaction process, we design a complementary representation supervised learning strategy (CRSL-Strategy). According to the similarities and differences of multisource features, it can adaptively promote complementary feature learning, thus generating a nonredundant multisource representation. The method has been verified to be effective on multiple RS datasets. The code is open source at: https://github.com/Xidian-AIGroup190726/Ping-Pie-I3RL-Net.git.**

***Index Terms*— Complementary representation, deep learning, fusion classification, multisource remote sensing, panchromatic (PAN) and multispectral (MS), self-attention mechanism.**

I. Introduction

With the rapid progress of remote sensing technology, in recent years, the types of various domestic satellites such as Gaofen satellites, communication satellites, and radars have been increasing. At the same time, the ways of obtaining image data have become more diverse [1]. We can easily obtain multisource data with complementary information, particularly panchromatic (PAN) data with rich spatial information and multispectral (MS) data with high spectral resolution [2]. Multisource remote sensing image classification technology is a very prominent field in the current field of remote sensing technology, which has important application value in urban planning, agricultural development, military, and other fields [3], [4], [5]. Its goal is to interpret the classes and distribution of ground objects based on their electromagnetic radiation characteristics [6].

There are classification methods based on single-source data [7], [8], [9], [10], [11]. For example, the hyperspectral image (HSI) contains high-level spectral details, which can be used to discriminate ground objects. Duan et al. [12] first constructed the structural outline of HSI, then sent it to the spectral classifier, and finally fused the class probabilities of the two stages. In [13], an edge-aware-based variational framework is constructed to extract the structural information of HSI. Zhao et al. [14] introduced a self-supervised method to solve the problem of few-sample labeling in HSI datasets. Huang et al. [15] mine the physical characteristics of SAR images and combine physical perception features with CNN to obtain a more comprehensive representation. Although the above methods have achieved good performance, the single-source data often contains incomplete information, which makes it difficult to accurately divide the region composed of pixels of the same category. The multisource classification methods exploit the complementarity characteristics between different sources to improve classification accuracy. For example, MS images are characterized by lower spatial resolution but higher spectral resolution, which makes it possible to identify land cover efficiently. PAN images have the characteristics of poor spectral resolution but strong spatial resolution, which can describe the outline and boundary of ground objects [16].

In recent years, multisource remote sensing image classification methods based on PAN and MS have emerged one after another. However, it is still a challenge to establish the relationships among multisource remote sensing images, make full use of data complementarity, and reduce information redundancy to improve the performance of classification models.

There are usually two types of strategies for multisource image classification tasks based on PAN and MS. The traditional methods focus on the fusion of multisource data in order to get fusion images of rich spatial and spectral features [17], [18]. For example, the component replacement (CS) method [19], [20] usually substitutes the spatial component of MS with PAN in the transformed spatial domain and then conducts the inverse transformation to obtain the fused HR-MS

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image. Frequently used CS methods include intensity hue saturation transform (IHS) [21], [22] and principal component analysis (PCA) [23], [24]. In the multiresolution analysis (MRA) methods [25], [26], wavelet transform [27] or Laplace pyramid transform [28] is used to decompose PAN and MS images to obtain multiscale and multiresolution images, and then we use a specific fusion strategy to fuse the multiresolution images. However, in practical applications, they all have some spectral distortion and spatial artifacts, which affect the fusion results.

In the deep learning approach, the features are extracted from multisource images, and then they are combined for classification [29], [30]. In [31], a three-branch spatial-channel progressive fusion ResNet (SCPF-ResNet) network is designed to enhance the spatial features of large targets and spectral features between channels. The third branch progressively strengthens the pure independent features extracted by PAN and MS branches. The three-branch network has more parameters and a heavier training burden than the two-branch network. Ma et al. [32] combine PCA and nonsubsampled contourlet transformation (NSCT) methods to adaptively transfer the dominant features of PAN and MS to each other. At the same time, they design an information progressive transfer fusion module to get the best-shared feature layer that is conducive to classification. Li et al. [33] solve the problem of small interclass differences and large intraclass differences inherent in remote sensing images from the perspective of metric learning. Wang et al. [34] first performed feature extraction on dual-source data. In the feature fusion stage, complementary information is adaptively fused by capturing the long-range dependencies between multisource features. In order to sort out multisource collaborative information and adjust the unbalanced distribution of features, Zhang et al. [35] used structural optimization as a guide to transmit spectral, spatial, and elevation information, effectively reducing the uncertainty of spectral information and solving the redundant problem of multisource information. Wang et al. [36] proposed a network to learn transferable multimodal feature representations through modal and semantic dual enhancement. At the same time, the knowledge learned by the previous network is retained to adjust the update of the classifier. In addition, in order to balance multisource information, a multibranch parameter interactive update strategy is adopted to optimize the network jointly.

The networks and algorithms designed in the above works have made good progress, encouraging us to further research multisource RS fusion classification tasks with deep learning methods. However, based on previous studies, there are still several problems worth exploring.

1) Due to the gap in the imaging mechanism, the underlying feature representations of multisource remote sensing data (such as PAN and MS) often have significant differences, resulting in a semantic gap between multisource high-level features, which is not conducive to the integration of multisource information. In addition, failure to strengthen single-source information will lead to single-source features failing to focus on more informative regions, which refers to feature-related foreground objects in the image.
2) Multisource remote sensing images describing the same scene usually have information redundancy during feature fusion, which makes the information useful for classification submerged in a large amount of redundant information.

In view of these problems, an I^3RL-Net is proposed to classify multisource RS images. The main contributions are summarized as follows.

1) We propose a cross-source interactive enhanced fusion module (CIEF-Module). First, it can focus on more informative regions by strengthening the feature dependencies within sources. Second, the semantic gap between the multisource high-level features can be effectively mitigated through the multilevel intersource interaction, which benefits the integration of multisource information.
2) To reduce the intersource information redundancy during feature fusion, we design a complementary representation supervised learning strategy (CRSL-Strategy) in the process of cross-source interaction. According to the similarities and differences between PAN and MS, an optimization loss function was established to retain the informative source's features adaptively, and another source filters out the complementary information to promote the complementary learning of multisource features.

The remainder of this article consists of the following contents. Section II reviews the work related to our designed approach. Section III describes our designs in detail. Section IV covers the introduction of the RS dataset used in our experiments and the analysis of experimental results. Last, this article's conclusion is in Section V.

II. Related Work

In this part, we discuss the work related to our designed approach. First, we briefly introduce the self-attention mechanism in transformer. Furthermore, we review the attention mechanism-based methods in multisource tasks and the related work of multisource representation learning.

A. Self-Attention in Transformer

Transformer have shown SOTA performance in natural language process [37], computer vision [38], [39], and other AI fields. The self-attention mechanism is a core component of the transformer model. As shown in Fig. 1, $\mathcal{X}$ is a feature of a sequence of input samples. Q, K, and V are obtained by mapping $\mathcal{X}$ through three randomly initialized matrices. Then we gain the output of the self-attention mechanism $\mathcal{X}'$ according to the following:

$$\mathcal{X}' = \text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V. \quad (1)$$

There are many studies based on the self-attention mechanism. In order to reduce the secondary computational complexity of self-attention and explore potential relationships between samples, Guo et al. [40] propose MLP external attention. Zhou et al. [41] apply the multihead self-attention mechanism to explore the connection between multiple temporal features, making full use of temporal information. The

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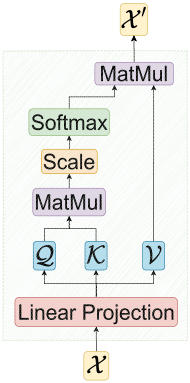


Fig. 1. Flowchart of the self-attention mechanism.

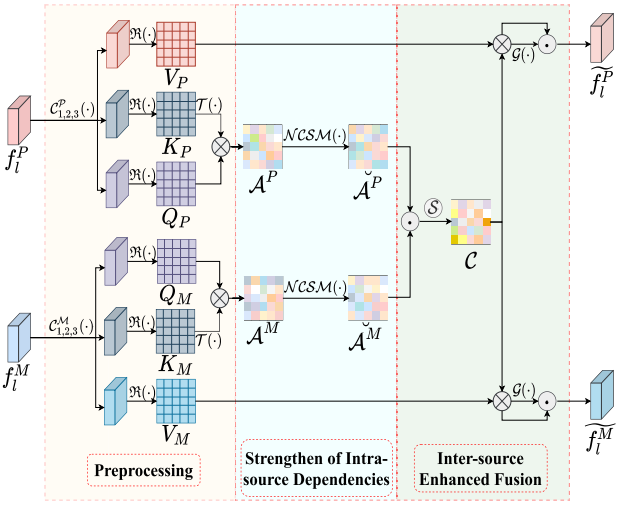


Fig. 2. Flowchart of CIEF-Module. $\mathcal{C}^{*}_{1,2,3}(\cdot)$ represents three parallel convolutional braches. $\tau(\cdot)$ stands for the transpose of a matrix. $\mathcal{G}(\cdot)$ refers to the gated weighting function.

self-attention mechanism has good global feature aggregation capabilities. Although research on self-attention mechanisms and downstream tasks have made good progress, when applied to remote-sensing images, attention is not focused enough on foreground objects, which can easily lead to interference in the recognition of pixels to be classified by other surrounding categories.

B. Multisource Representation Learning

A common fundamental problem in multisource tasks is learning how to represent and summarize multisource data based on the complementarity and redundancy of multiple sources. The heterogeneity of multisource data makes it challenging to construct such a representation. Multisource representation means eliminating information redundancy by utilizing the complementarity between multiple sources to learn better feature representation [42]. Geng et al. [43] proposes an information bottleneck fusion network for multisource image fusion to integrate multisource information to obtain a comprehensive multisource representation effectively. In [44], an interpretable classification model is proposed. In order to take full advantage of the correlation of multisource data, it aligns spatial and spectral information at the end of the network to jointly represent multisource information. Among them, the dictionary distribution is aligned so that the complementary information of multisource data can guide more accurate classification.

Considering the heterogeneity of multisource remote sensing data, most existing algorithms don't fully explore the similarities and differences between multiple sources, resulting in semantic gaps and information redundancy. Therefore, these methods are difficult to directly apply to multisource data, such as PAN and MS data.

III. Method

Our proposed I^3RL-Net consists of two parts: a CIEF-Module and a CRSL-Strategy. Their details are described in the following contents.

A. Cross-Source Interactive Enhanced Fusion Module

Existing cross-source interaction methods usually focus on intrasource or intersource relationships separately, which makes multisource information integration difficult. In particular, we argue that intrasource relations are a further complement to intersource relations [45]. Based on this, in the feature extraction stage, we propose a CIEF-Module, as shown in Fig. 2, which facilitates the interaction between the multilevel features from multiple sources to effectively alleviate the intersource semantic gap. It establishes a connection between intrasource and intersource relations by strengthening the internal dependencies of single-source features and learning the correlation between multisource features. We elaborate on the designed details as follows.

1) Preprocessing: We first use several residual modules to extract multilevel PAN and MS features and get $f_l^P \in \mathbb{R}^{4H\times 4W\times C_l}$, $f_l^M \in \mathbb{R}^{H\times W\times C_l}$, respectively. Since the self-attention mechanism in the transformer model can utilize the inherent information within the features for interaction, we use it to model the intrasource feature global dependencies. $f_l^P \in \mathbb{R}^{4H\times 4W\times C_l}$ is taken as an example to illustrate the calculation process, and we get the self-correlation matrix $\mathcal{A}^P \in \mathbb{R}^{HW\times HW\times 1}$ as follows:

$$\begin{cases} Q_P = \Re\left(\mathcal{C}_1^P\left(\mathcal{D}_{\text{avg}}\left(f_l^P\right)\right)\right) \\ K_P = \Re\left(\mathcal{C}_2^P\left(\mathcal{D}_{\text{avg}}\left(f_l^P\right)\right)\right) \\ V_P = \Re\left(\mathcal{C}_3^P\left(\mathcal{D}_{\text{avg}}\left(f_l^P\right)\right)\right) \\ \mathcal{A}^P = \text{L2Norm}\left(Q_P \cdot (K_P)^T/\sqrt{d_k}\right) \end{cases} \tag{2}$$

where $\mathcal{D}_{\text{avg}}(\cdot)$ reduces the resolution of f_l^P to be the same as f_l^M. $\mathcal{C}^P_{1,2,3}(\cdot)$ are three parallel convolutional layers with 1×1 kernel size, and $\Re(\cdot)$ is used to change the size of f_l^P from $\mathbb{R}^{H\times W\times C_l}$ to $\mathbb{R}^{HW\times C_l}$.

At the same time, we perform the same operation as above for f_l^M and get $\mathcal{A}^M \in \mathbb{R}^{HW\times HW\times 1}$, except for the downsampling operation $\mathcal{D}_{\text{avg}}(\cdot)$.

2) Strengthen of Intra-Source Dependencies: In order to refine single-source relevant features and focus on more informative regions, our goal is to pay attention to the most relevant information within the single source. However, since the self-attention mechanism lacks such concentration, it does not emphasize the degree of correlation between features. Thereby, it is easily distracted. Thus, we design a neighborhood consensus self-matching operation ($\mathcal{NCSM}(\cdot)$), which can emphasize the degree of correlation between row features and column features within the single source, as shown in Fig. 3.

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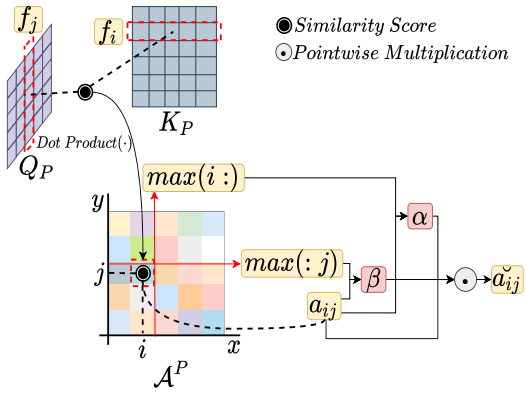


Fig. 3. Schematic of the $\mathcal{NCSM}(\cdot)$ operation. f_i refers to the ith row features, and f_j refers to the jth column features.

Specifically, after obtaining $\mathcal{A}^P \in \mathbb{R}^{HW\times HW\times 1}$ and $\mathcal{A}^M \in \mathbb{R}^{HW\times HW\times 1}$, we multiply each element of the self-correlation matrix by two coefficients $\alpha, \beta \in (0, 1)$ separately and then get $\breve{\mathcal{A}}^P$ and $\breve{\mathcal{A}}^M$. Taking $\mathcal{A}^P$ as an example, we get $\breve{\mathcal{A}}^P$ according to the following calculation:

$$\breve{a}_{ij} = \mathcal{NCSM}(a_{ij}) = \alpha \cdot \beta \cdot a_{ij}$$
$$\text{s.t.} \begin{cases} \alpha = \dfrac{a_{ij}}{\max(i:)} \\ \beta = \dfrac{a_{ij}}{\max(:j)} \\ a_{ij} \in \mathcal{A}^P \end{cases} \tag{3}$$

where a_{ij} represents the correlation of feature pair (f_i, f_j). f_i represents the ith row feature, and f_j represents the jth column feature. α and β are the ratios of the similarity score of the particular match (f_i, f_j) with the highest similarity scores for row i and column j in $\mathcal{A}^P$.

Similarly, we get $\mathcal{A}^M$. This neighborhood consensus self-matching operation is designed to add mutual nearest neighbor constraint-liked to each matching score a_{ij}, that is to say, among all matched features, if and only if the highest similarity $\max(i:)$ to the feature f_i is f_j, and if the highest similarity $\max(:j)$ to the feature f_j is f_i, the weight coefficient obtained α and β is closer to 1. Otherwise, it is closer to 0. In other words, α and β reflect the matching degree between their features, and we multiply them on $\mathcal{A}^P$ and $\mathcal{A}^M$ to emphasize the strength of correlation between features. It can make similar features more similar and dissimilar features less similar. As a result, the score of mutual nearest neighbor matching is increased, while the score of nonmutual nearest neighbor matching is decreased, so more attention is paid to intrasource related features.

3) Inter-Source Enhanced Fusion: After the above operations, the intrasource relevant information is strengthened, then the intersource fusion operation is performed in two self-correlation matrices as follows:

$$\mathcal{C} = \sigma\left(\breve{\mathcal{A}}^P \odot \breve{\mathcal{A}}^M\right) \tag{4}$$

where $\odot$ refers to pointwise multiplication, and σ represents the Softmax function.

The matrix $\mathcal{C} \in \mathbb{R}^{HW\times HW\times 1}$ reflects the cross-source correlation. Moreover, it incorporates the most relevant feature information of the two sources. Further, we carry out implicit feature transfer through this matrix to achieve bidirectional feature enhancement between multiple sources. At the same time, considering the adaptability of transitive features to potential changes in appearance information, such as noise, we gate-weighted each value of the transitive features, which is used to determine how much information in the transmitted features will be received. This process can be expressed as follows:

$$\begin{cases} \left[\widetilde{f_l^P}\right] = (\mathcal{C} \times V_P) \odot \delta\left(\mathcal{C} \times V_P; w_p\right) \\ \left[\widetilde{f_l^M}\right] = (\mathcal{C} \times V_M) \odot \delta(\mathcal{C} \times V_M; w_m) \end{cases} \tag{5}$$

where $\odot$ stands for pointwise multiplication, same as before. δ represents the Sigmoid function. w_p and w_m are the learnable parameters. "$\times$" represents the matrix multiplication operation.

Then the bidirectional enhanced features $[\widetilde{f_l^P}]$ and $[\widetilde{f_l^M}]$ which contain each other's most related feature information are obtained, and they are added with the input of the CIEF-Module to continue the following feature extraction.

In summary, through multilevel cross-source feature interaction, intrasource relevant features are strengthened and transferred between multisources. The multisource features are well complemented, and the semantic gap between the multisource high-level features is reduced, which benefits the integration of multisource information.

B. Complementary Representation Supervised Learning Strategy

Most existing algorithms usually use direct fusion or adaptive feature fusion based on the attention mechanism for features after the interaction, ignoring that there is intersource information redundancy during multisource interaction, which is not conducive to network mining useful information for classification.

To solve this problem, as shown in Fig. 4, we propose a CRSL-Strategy. According to the similarities and differences between the features of multiple sources, an optimization loss function is established to adaptively enhance the learning of complementary features of the two sources to reduce inter-source information redundancy. We elaborate on the details of the design in the following.

It is generally believed that the V component of the self-attention mechanism can represent the content information of single-source features. Therefore, we establish an optimization for $V_P \in \mathbb{R}^{H\times W\times C_l}$ and $V_M \in \mathbb{R}^{H\times W\times C_l}$ as follows to make them tend to express complementary features:

$$\min \mathcal{L}_p = \gamma_p \cdot \|V_P\|_2 + \gamma_m \cdot \|V_M\|_2$$
$$\text{s.t.} \begin{cases} \gamma_p = \dfrac{1}{1 + e^{\kappa\cdot\xi}} \\ \gamma_m = 1 - \gamma_p \end{cases} \tag{6}$$

where $\|\cdot\|_2$ refers to $L2$ norm. The form of the Sigmoid function is used to define coefficient $\gamma_p \in (0, 1)$ and $\gamma_m \in (0, 1)$ as shown above.

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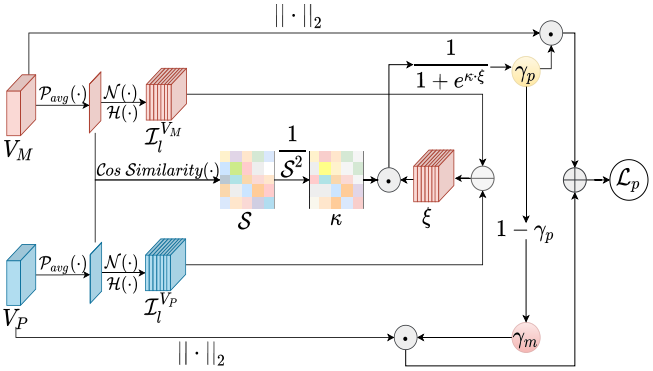


Fig. 4. Schematic of the computational procedure of the CRSL-Strategy.

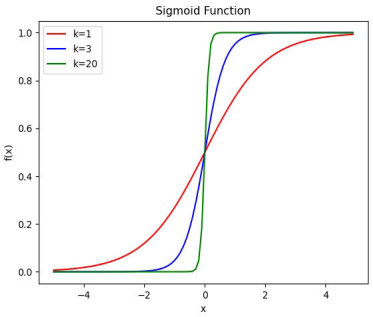


Fig. 5. Curve of the Sigmoid function $f(x) = 1/(1+e^{-kx})$. We can see that the larger k is, the faster the function goes down.

1) Design of κ: In order to preserve the single source's specific information which refers to the difference information compared to another source and avoid similar features appearing in multiple sources at the same time, we define κ based on the cosine similarity between V_P and V_M as follows:

$$\kappa = \frac{1}{\mathcal{S}^2(V_P, V_M)} \tag{7}$$

$$\mathcal{S}(V_P, V_M) = \frac{\mathcal{P}_{\text{avg}}(V_P) \cdot \mathcal{P}_{\text{avg}}(V_M)}{\left\|\mathcal{P}_{\text{avg}}(V_P)\right\|_2 \cdot \left\|\mathcal{P}_{\text{avg}}(V_M)\right\|_2} \tag{8}$$

where $\mathcal{P}_{\text{avg}}(\cdot)$ is the global average pooling of channels to focus on the information of each channel and $\|\cdot\|_2$ denotes L2 norm.

2) Design of ξ: To retain the features of the informative source, ξ is defined by the differences in feature richness between PAN and MS. Considering that information entropy [46] can be used to represent the amount of information that something contains, we use it to represent the richness of features. For $V \in \mathbb{R}^{H\times W\times C_l}$ components of PAN and MS features, we do the following operation, respectively, and get $\mathcal{I}_l^{V_P} \in \mathbb{R}^{1\times 1\times C_l}$ and $\mathcal{I}_l^{V_M} \in \mathbb{R}^{1\times 1\times C_l}$:

$$V_*' = \mathcal{N}\left(\mathcal{P}_{\text{avg}}(V_*)\right) \tag{9}$$

where $*$ stands for P or M. We use max-min normalization $\mathcal{N}(\cdot)$ operation to adjust the values of V_* for paying attention to the feature distribution of V_*. Then calculate the information entropy $\mathcal{I}_l^{V_*} \in \mathbb{R}^{1\times 1\times C_l}$ as follows:

$$\mathcal{I}_l^{V_*} = \mathcal{H}\left(V_*'\right) \tag{10}$$

where $\mathcal{H}(\cdot)$ stands for information entropy calculation function. And given the content component V_*' of single-source features, the information entropy calculation process is defined as follows:

$$\mathcal{H}\left(V_*'\right) = -\sum_{i=0}^{v} p_i \log p_i \tag{11}$$

where p_i represents the probability that the value i appears in V_*'.

Then ξ is defined as follows:

$$\xi = \mathcal{I}_l^{V_P} - \mathcal{I}_l^{V_M}. \tag{12}$$

From the above equation, we can see that when κ is large, that is, the similarity between V_P and V_M is low, at the same time, if $\xi > 0$, that is, PAN features are richer than MS features. So according to (6), the calculated γ_p is closer to 0, but γ_m is closer to 1. This means that we punish MS features more so that they retain the complementary features to avoid more information similar to PAN features appearing in MS features. And vice versa. Therefore, the information redundancy between PAN and MS features is mitigated. From the perspective of the Sigmoid function $f(x) = 1/(1+e^{-kx})$, as shown in Fig. 5, the larger the value of k, the faster the function goes down, resulting in the weight distribution tends to be unbalanced. In addition, if κ is small, that is to say, the features of PAN and MS are similar to each other, and their information richness is also similar. So κ is going to approach 1, γ_p and γ_m are approaching 0.5. As a result, the degree of punishment for PAN and MS features is close.

In general, through the complementary representation supervised learning of V_P and V_M, features of the informative source are retained, and information redundancy between multiple sources is reduced.

C. Construction of the I^3RL-Net

In this section, we construct the designed I^3RL-Net as a whole. As shown in Fig. 6, the structure of I^3RL-Net consists of the following parts.

1) Feature Extraction Based on Residual Blocks: PAN and MS image patches are transferred to multiple residual modules with the same structure but unshared weights to extract multilevel features $f_l^P \in \mathbb{R}^{4H\times 4W\times C_l}$, $f_l^M \in \mathbb{R}^{H\times W\times C_l}$.

2) CIEF-Module: Given the paired single-source features f_l^P and f_l^M at the same level, the CIEF-Module is used to facilitate the interaction between multiple sources in the process of feature generation. It can be represented as follows:

$$\widetilde{f_l^P}, \widetilde{f_l^M} = \text{CIEF}_l\left(f_l^P, f_l^M\right). \tag{13}$$

After the interaction, they are added to the input features f_l^P, f_l^M and sent to the double-branch network to continue the following feature extraction.

3) Feature Fusion and Classification: For high-level single-source features $\widetilde{f_L^P}$ and $\widetilde{f_L^M}$, which have a small semantic gap, we do as follows:

$$f^{\text{fusion}} = \mathcal{C}\text{at}\left[\mathcal{P}_{\text{avg}}\left(f_L^P + \widetilde{f_L^P}\right), \mathcal{P}_{\text{avg}}\left(f_L^M + \widetilde{f_L^M}\right)\right]. \tag{14}$$

$\mathcal{C}\text{at}(\cdot)$ represents the cascade of channel dimension. Then, we transform the fusion feature f^{fusion} into a vector for the

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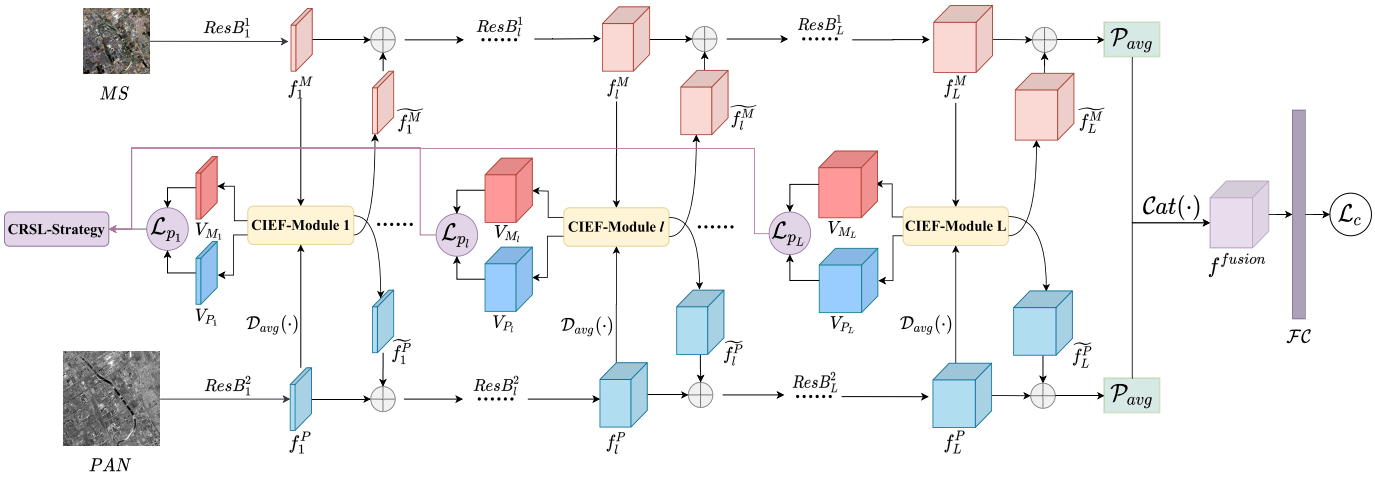


Fig. 6. Network structure of our designed I^3RL-Net. The PAN and MS image patches are first sent to several residual modules to extract multilevel features $f_l^P \in \mathbb{R}^{4H\times 4W\times C_l}$ and $f_l^M \in \mathbb{R}^{H\times W\times C_l}$. Then, they interact through multiple CIEF-Modules to narrow the semantic gap. In the process of interaction, the CRSL-Strategy is used to reduce intersource information redundancy. Finally, we concatenate the features obtained from the dual branches in channel dimension to obtain multisource high-level feature representations f^{fusion} for classification.

final predictive classification probability output. The total loss of the network is defined as follows:

$$\mathcal{L}_t = \mathcal{L}_c + \lambda \cdot \sum_{l=1}^{L} \mathcal{L}_{p_l} \tag{15}$$

where $\mathcal{L}_t$ refers to the total loss, $\mathcal{L}_c$ represents the cross entropy loss and $\mathcal{L}_{p_l}$ represents the lth feature level's complementary representation supervised learning loss. To keep the same order of magnitude as $\mathcal{L}_c$, we perform the normalization operation on $\mathcal{L}_{p_l}$. The value of λ and the number of CIEF-Module l are analyzed in the experimental part.

IV. Experiments

In this part, we introduce the four multisource RS datasets in detail. Next, we provide our experimental setup and conduct a series of experiments to verify the performance improvement of our design for classification tasks.

A. Dataset Description

The four PAN and MS datasets we used were purchased from GF Research Institute and annotated by our laboratory. The images are all obtained by the Gaofen-2 satellite. Their details are as follows.

1) Hohhot Dataset: As shown in Fig. 7(a), MS image is a color image consisting of R, G, B, and NIR four channels with 2001 × 2101 × 4 pixels, and its spatial resolution is 4 m. PAN image is a single-channel grayscale image containing 8004 × 8404 × 1 pixels, and its spatial resolution is 1 m. As we can see from the GT of the Hohhot image, there are 11 kinds of labeled ground objects, including water, road, bare land, vegetation, farmland, and six types of building areas.

2) Xi'an Dataset: As shown in Fig. 7(b), MS image is a color image consisting of 4548 × 4544 × 4 pixels, and its spatial resolution is 4 m. PAN image is a grayscale image containing 18 192 × 18 164 × 1 pixels, and its spatial resolution is 1 m. As we can see from the GT of Xi'an dataset, it contains water, two types of roads, bare land, farmland, low vegetation, tree, and five kinds of building areas.

3) Beijing Dataset: As shown in Fig. 7(c), MS image is a color image consisting of 6908 × 7300 × 4 pixels, and its spatial resolution is 4 m. PAN image is a grayscale image containing 27 632 × 29 200 × 1 pixels, and its spatial resolution is 1 m. As we can see from the GT of Beijing dataset, there are ten types of labeled ground objects. They are water, road, bare land, two types of vegetation, and five types of building areas.

4) Shanghai Dataset: As shown in Fig. 7(d), MS image is a color image consisting of 6908 × 7300 × 4 pixels, and its spatial resolution is 4 m. PAN image is a grayscale image containing 27 632 × 29 200 × 1 pixels, and its spatial resolution is 1 m. We can see from the GT of Shanghai dataset that it contains ten classes of labeled ground objects such as water, road, bare land, two kinds of vegetation, and five kinds of buildings.

All experiments in this article run on the system with RTX2080Ti 11-GB GPU and 128-GB RAM under Ubuntu 20.04 LTS. Our designed model is trained based on the PyTorch deep learning framework.

B. Experimental Setup

Before network training, we crop the input MS image to the size of 16 × 16 × 4 pixels, and since the resolution ratio of MS and PAN images is 1:4, the corresponding PAN image is cropped to the size of 64 × 64 × 1 pixels. We randomly select about 2% samples of each ground object category for training. For better training, the intersection union ratio (IOU) of the training samples and the testing samples is less than 1/3. In our experiments, the initial learning rate is 0.001, the training epochs are 30, and the batch size of training samples is 48. The details of the parameter configuration of our designed network are shown in Table I.

In this article, we measure the performance of the designed model using the following three metrics: overall accuracy (OA)—the percentage of correctly classified samples; average accuracy (AA)—the mean of classification accuracy; Kappa

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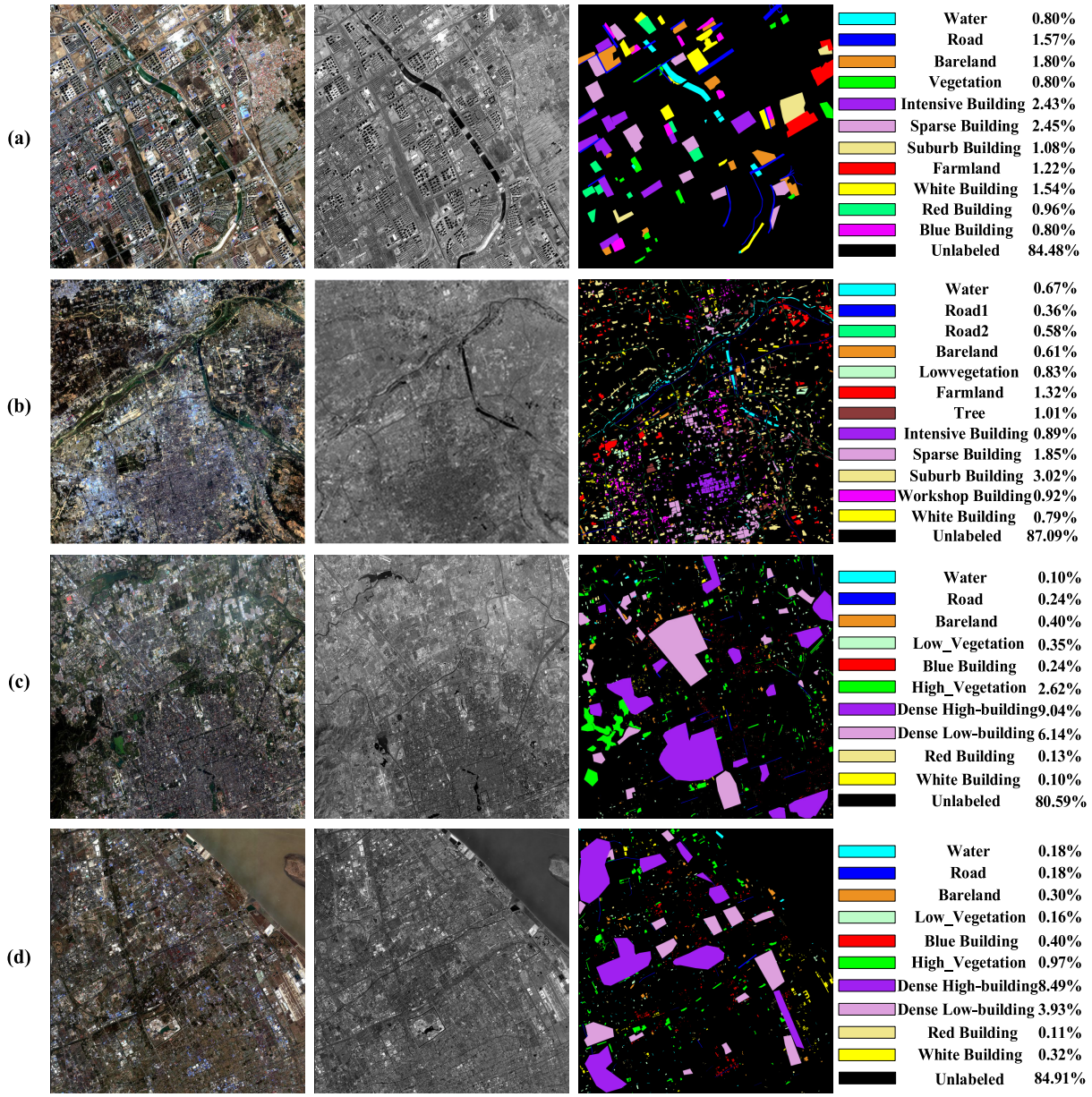


Fig. 7. (a) Images of the Hohhot dataset. (b) Images of the Xi'an dataset. (c) Images of the Beijing dataset. (d) Images of the Shanghai dataset. The columns from left to right are MS images, PAN images, GT images, category labels, and their proportions corresponding to the GT images. For the convenience of the display, we reduce the PAN image to the same size as the MS image.

coefficient (Kappa)—provides mutual information about the substantial agreement between GT plots and classification plots. By conducting five experiments, the experimental results are obtained and subsequently averaged to derive the final value.

C. Selection and Analysis of Hyperparameters

In this section, we compare and analyze the values of the two hyperparameters of the network and take OA as the metric.

1) λ in (15): The weight of CRSL-Strategy. It controls how much this strategy affects the output of the network. We start the value of λ from 0.1 with a step size of 0.1 and take ten values. In particular, we take two additional values near the optimal value for each dataset. As shown in Table II, on the Hohhot and Xi'an datasets, the best classification performance is obtained when $\lambda = 0.5$, and when λ is far from 0.5, the classification performance decreases but to a small extent. This shows that on these two datasets, the choice of λ has relatively little impact on the results. However, for the Beijing and Shanghai datasets, the situation is slightly different. Since these two datasets are relatively complex, CRSL-Strategy can reduce a large amount of information redundancy between sources and retain the unique information of a single source, thereby filtering out information useful for classification. Therefore, we can observe from Table II that when $\lambda = 0.3$, they achieve the best classification results. When λ is far away from 0.3, the OA value shows large fluctuations. Through these experimental results, we can conclude that for different datasets, the value of the optimal

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TABLE I
PARAMETER CONFIGURATION OF OUR NETWORK

Phase	Input	PAN_Branch	MS_Branch	Output
Prejust	$64 \times 64 \times 1$ $16 \times 16 \times 4$	$\begin{bmatrix} Conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix}, 1 \rightarrow 64$	$\begin{bmatrix} Conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix}, 4 \rightarrow 64$	$64 \times 64 \times 64$ $16 \times 16 \times 64$
Feature_Extraction	$64 \times 64 \times 64$ $16 \times 16 \times 64$	$ResB_1^1$: 3×3, $64 \rightarrow 64$, *downsample* $ResB_2^1$: 3×3, $64 \rightarrow 128$, *downsample* $ResB_3^1$: 3×3, $128 \rightarrow 256$, *downsample* $ResB_4^1$: 3×3, $256 \rightarrow 512$, *downsample*	$ResB_1^2$: 3×3, $64 \rightarrow 64$ $ResB_2^2$: 3×3, $64 \rightarrow 128$ $ResB_3^2$: 3×3, $128 \rightarrow 256$ $ResB_4^2$: 3×3, $256 \rightarrow 512$	$16 \times 16 \times 64$ $16 \times 16 \times 128$ $16 \times 16 \times 256$ $16 \times 16 \times 512$
CIEF-Module	$16 \times 16 \times 64$ $16 \times 16 \times 128$ $16 \times 16 \times 256$ $16 \times 16 \times 512$	$16 \times 16 \times C_l \rightarrow 256 \times 256$ $256 \times 256 \rightarrow 256 \times 256$ $256 \times 256 \rightarrow 16 \times 16 \times C_l$ $(l = 1, 2, 3, 4)$	$16 \times 16 \times C_l \rightarrow 256 \times 256$ $256 \times 256 \rightarrow 256 \times 256$ $256 \times 256 \rightarrow 16 \times 16 \times C_l$ $(l = 1, 2, 3, 4)$	$16 \times 16 \times 64, loss1$ $16 \times 16 \times 128, loss2$ $16 \times 16 \times 256, loss3$ $16 \times 16 \times 512, loss4$
Feature_Fusion	$16 \times 16 \times 512$ $16 \times 16 \times 512$	*Global_Average_Pooling*	*Global_Average_Pooling*	$1 \times 1 \times 512$ $1 \times 1 \times 512$
Classification	$1 \times 1 \times 512$ $1 \times 1 \times 512$	$Concat: \begin{bmatrix} 1\times 1\times 512 \\ 1\times 1\times 512 \end{bmatrix} \rightarrow \begin{bmatrix} 1024, classes \end{bmatrix}$ $FC: \begin{bmatrix} 1024, classes \end{bmatrix}$		*classes*

TABLE II
EXPERIMENTAL RESULTS OF HYPERPARAMETER COMPARISON

	λ	Hohhot	Xi'an	Beijing	Shanghai
OA(%)	0.1	99.05	99.03	94.14	91.78
	0.2	99.09	99.07	95.01	93.10
	0.25	—	—	95.54	94.68
	0.3	99.14	99.13	**95.74**	**95.76**
	0.35	—	—	95.61	95.58
	0.4	99.19	99.17	95.52	95.49
	0.45	99.21	99.20	—	—
	0.5	**99.24**	**99.25**	95.33	95.21
	0.55	99.21	99.21	—	—
	0.6	99.20	99.18	95.29	95.17
	0.7	99.18	99.16	95.25	95.14
	0.8	99.16	99.14	92.84	94.12
	0.9	99.13	99.13	90.73	91.23

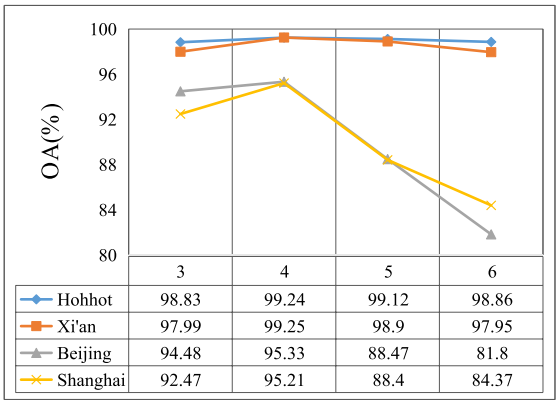


Fig. 8. Evaluation of the value of hyperparameter l. We set the values of l to 3–6 for our experiments.

λ will change. This further validates the robustness of our method, as it is able to adaptively select optimal λ' value based on the characteristics of the dataset.

2) l in (15): The number of CIEF-Module. It reflects the degree of multisource interaction. We conduct experiments with hyperparameter l by fixing the value of hyperparameter λ to 0.5, as shown in Fig. 8. When $l = 4$, the network classification accuracy is the highest. This is because when l is small, it is difficult to carry out effective and adequate multisource interaction, and the problem of the intersource semantic gap can't be well resolved. Moreover, the network is clunky when l is large. And with the help of the CIEF-Module, the deep feature information of PAN and MS is more and more similar, which leads to more intersource redundant information and is not conducive to classification.

D. Module Comparison Experiments

In this part, the Hohhot dataset is used for the analysis of module comparison experiments. The designed CIEF-Module is based on self-attention in transformer, so we first compare the classification performance of our design SA + NCSM with using pure SA. Then, we compare the designed CIEF-Module with our corresponding base module that performs multisource fusion via cross-attention on PAN and MS features extracted through the residual block. We directly add these modules to the baseline to verify the effectiveness of our design better.

1) SA + NCSM Versus SA: We can see from Table III that the values of the three classification metrics of SA + NCSM are higher than SA. This can be explained by the fact that

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TABLE III
MODULE COMPARISON EXPERIMENT RESULTS ON HOHHOT DATASET

Classes	SA+NCSM	SA	CIEF-Module	Base-Module
c_1:Water	97.39	98.79	99.09	99.14
c_2:Road	95.20	92.85	95.96	95.75
c_3:Bareland	96.54	92.50	95.47	96.33
c_4:Vegetation	96.52	94.95	97.26	94.40
c_5:Intensive Building	95.89	91.73	97.89	95.17
c_6:Sparse Building	90.38	94.95	98.74	94.22
c_7:Suburb Building	95.60	95.96	99.07	96.25
c_8:Farmland	96.73	96.18	87.98	97.70
c_9:White Building	94.92	91.28	98.49	93.55
c_{10}:Red Building	94.54	92.82	93.88	88.19
c_{11}:Blue Building	91.09	93.23	91.33	92.44
OA(%)	94.75	93.73	96.34	94.92
AA(%)	94.98	94.11	95.92	94.83
Kappa(%)	94.12	92.98	95.91	94.31

TABLE IV
ABLATION STUDY RESULTS OF EACH MODULE ON HOHHOT DATASET

Baseline	CIEF	CRSL	OA(%)	AA(%)	Kappa(%)
✓			91.72	91.28	90.72
✓	✓		96.34	95.92	95.91
✓		✓	97.32	96.91	97.00
✓	✓	✓	99.24	99.26	99.15

due to the difference in the distribution of objects in remote sensing images, focusing on strongly correlated features can make the network focus more on features with rich information. However, pure SA can only simply extract relevant information from itself without emphasizing the degree of correlation between features. NCSM assigns a weight to each correlation value according to mutual nearest neighbor constraint liked to strengthen or weaken this correlation, so the combination of SA + NCSM achieves superior performance.

2) CIEF-Module Versus Base-Module: From Table III, the classification results of the CIEF-Module are better than the corresponding base version on the multisource RS datasets. The base version applies cross-attention to perform multisource fusion on PAN and MS features. Although this can achieve good classification performance, single-source features are not refined, and the problem of semantic feature differences in multisource data has not been well resolved. CIEF-Module exhibits competitive performance by gradually reducing the semantic gap between the multisource high-level features through multilevel feature interactions.

E. Ablation Study

This part studies the effects and contributions of designed modules and strategies on the Hohhot dataset. We use ResNet18 as the baseline network, simultaneously combining CIEF-Module and CRSL-Strategy for the ablation study. In particular, we maintain the same hyperparameters and experimental settings, then derive the results in Table IV.

We can see from Table IV that the OA of baseline ∪ CIEF-Module is 4.62% higher than baseline. Due to the semantic gap between the multisource high-level features, the CIEF-Module improves the classification accuracy. In addition, the contribution of CRSL-Strategy is larger due to the intersource information redundancy, which is up 5.6% compared to the baseline. The combination of baseline ∪ CIEF-Module ∪ CRSL-Strategy achieves the performance of SOTA, and its OA is 7.52% higher than the baseline. This is due to the fact that CRSL-Strategy reduces the information redundancy during cross-source interaction.

F. Comparison With Other Methods

We compare the performance of the designed I^3RL-Net with several existing SOTA models, including AHF-Net: the adaptive hybrid fusion network [47], SCPF-ResNet: the spatial-channel progressive fusion residual network [31], MPCA-Net: the multiscale progressive collaborative attention network [48], CRHFF: the cross-resolution hidden layer feature fusion approach [4], STGC-Net: the selectable transformer and Gist CNN network [49]. The details of the experiment are kept the same as in the original article.

1) Comparison on Hohhot Dataset: Table V and Fig. 9 present the experimental comparison and visualization results of our method on the Hohhot image.

AHF-Net first fuses multisource data and then combines feature fusion strategies to improve the network's ability to identify fused features. Therefore, it achieves decent classification results in various categories. However, the classification performance of AHF-Net is lower than that of other models due to not paying attention to the attribute features of multisource images.

Compared with AHF-Net, SCPF-ResNet has improved classification accuracy in most categories, especially c_2:Road has the largest improvement. It designs an adaptive spatial and channel attention module to strengthen the spatial and spectral features of the objects, thus improving the classification results of c_1:Water, c_2:Road, c_3:Bareland and other ground objects with prominent spectral and spatial features.

We can see from Table V that MPCA-Net achieves top results on c_{11}:Bluebuilding. This type of object is widely distributed and has high sample diversity. MPCA-Net's ADR-SS strategy is good at solving the problem of scale differences in category regions, so the classification accuracy of this category has been improved.

Compared with the above models, the results of CRHFF and STGC-Net on the indicators of OA, AA, and Kappa have been further improved. STGC-Net achieves the highest classification accuracy on c_2:Road, c_7:Suburb Building, c_{10}:Red Building. What these categories have in common is that the shape is usually relatively regular, but the distribution of roads is narrow, and the distribution of buildings is relatively spacious. STGC-Net proposes a subspace similarity recombination module (SSR-Module), which extracts singular and shared information separately while alleviating redundant information.

It is worth noting that compared with previously proposed methods, our designed I^3RL-Net achieves the best OA, AA, Kappa, and the best classification results in most categories. Especially c_3:Bareland, c_6:Sparse Building, c_8:Farmland have the most remarkable improvement. What these categories of objects have in common is that the distribution is relatively sparse and extensive. Based on the CRSL strategy, I^3RL-Net reduces the redundant features between objects, which is

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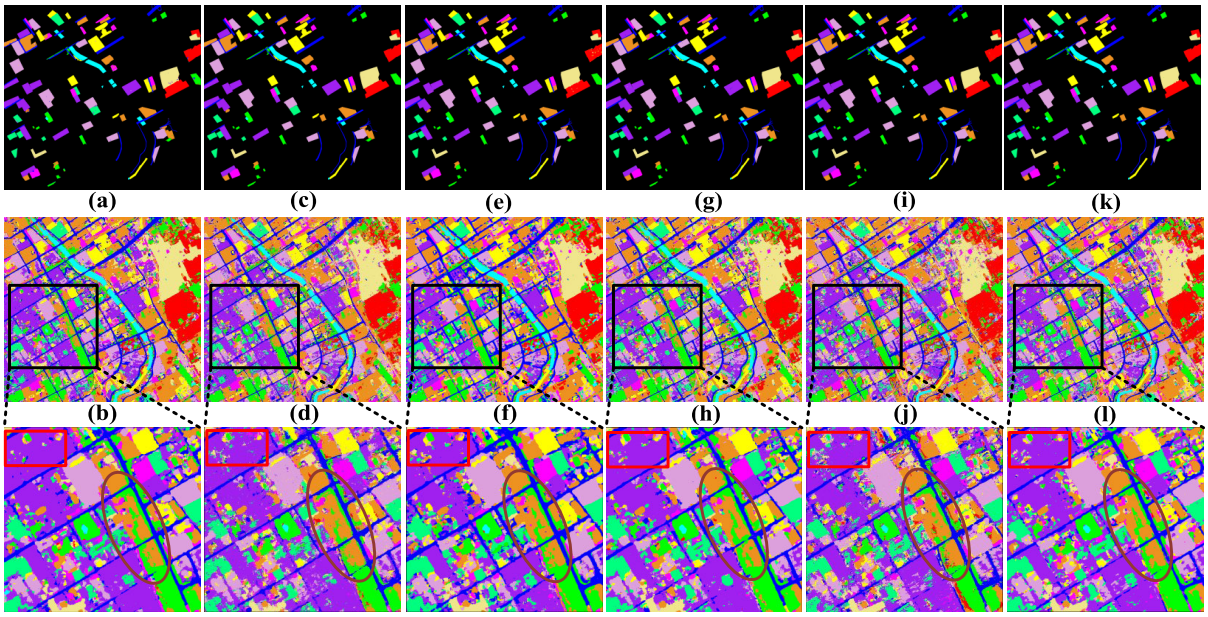


Fig. 9. Classification maps of different models on Hohhot dataset. The left column is GT, and the right column is the overall image. (a) and (b) AHF-Net. (c) and (d) SCPF-ResNet. (e) and (f) MPCA-Net. (g) and (h) CRHFF. (i) and (j) STGC-Net. (k) and (l) I^3RL-Net.

TABLE V
QUANTITATIVE RESULTS ON HOHHOT DATASET

Classes	AHF-Net	SCPF-ResNet	MPCA-Net	CRHFF	STGC-Net	OURS
c_1	99.22±0.18	99.69±0,16	99.61±0.09	98.56±0.07	99.42±0.11	**99.81±0.04**
c_2	94.35±0.34	97.21±0.12	96.95±0.11	96.98±0.12	**99.54±0.14**	98.24±0.30
c_3	96.25±0.32	97.30±0.25	96.13±0.15	98.31±0.18	98.95±0.23	**99.16±0.27**
c_4	96.81±0.29	98.48±0.11	97.63±0.19	98.47±0.21	99.29±0.10	**99.32±0.09**
c_5	98.52±0.36	97.53±0.28	97.84±0.09	97.94±0.08	99.20±0.31	**99.28±0.20**
c_6	97.23±0.18	97.64±0.07	96.20±0.27	96.99±0.11	98.89±0.24	**99.53±0.14**
c_7	98.10±0.27	98.54±0.23	98.88±0.31	99.03±0.19	**99.91±0.17**	99.19±0.13
c_8	97.83±0.31	98.65±0.18	97.47±0.24	98.80±0.20	99.37±0.13	**99.81±0.22**
c_9	96.22±0.35	97.96±0.13	96.78±0.14	98.26±0.11	98.91±0.11	**99.08±0.17**
c_{10}	96.15±0.27	97.61±0.14	96.79±0.17	97.93±0.16	**99.45±0.15**	99.22±0.26
c_{11}	95.69±0.25	97.63±0.21	**99.35±0.19**	97.35±0.23	98.94±0.16	99.28±0.15
OA(%)	96.82±0.19	97.86±0.13	97.38±0.12	98.24±0.12	99.23±0.11	**99.24±0.15**
AA(%)	96.64±0.14	97.78±0.14	97.60±0.11	98.06±0.08	99.20±0.13	**99.26±0.06**
Kappa(%)	96.27±0.13	97.61±0.16	97.27±0.15	98.09±0.10	99.14±0.09	**99.15±0.14**
Time(s)	412.18±28.84	222.23±20.65	542.28±29.89	431.79±25.96	**121.99±18.97**	505.02±27.04

more conducive to recognizing these open-ground objects. In addition, the design of SA + NCSM is more helpful in finding relevant small object information in open areas and can identify interclass differential features. Our model is weaker than STGC-Net and MPCA-Net on c_{10}: Red Building, c_{11}: Blue Building, but the difference is insignificant. On c_2:Road, c_7:Suburb Building, the classification precision of I^3RL-Net is not as good as STGC-Net. This may be because our model does not recognize the spatial features of road and suburban buildings well, while the multistream Gist spatial feature extraction module proposed by STGC-Net introduces the positional relationship between local features.

2) Comparison on Xi'an Dataset: The experimental results on the Xi'an dataset are presented in Table VI and Fig. 10. For the Xi'an dataset, our model I^3RL-Net achieves the highest classification results overall. In particular, the accuracy of c_3:Road2, c_5:Lowvegetation, c_8:Intensive Building, and c_9:Sparse Building has been significantly improved. The features of these categories are not similar, and I^3RL-Net emphasizes reducing the redundant information of inter-source features to increase the diversity of features. SCPF-ResNet achieves the highest accuracy on c7:Tree because trees are usually borderless and are easily confused with other categories, thus misclassifying them as close categories. In addition, STGC-Net achieves good classification accuracy on c_4:Bareland by introducing the positional relationship between local features. However, compared with our model, the classification accuracy improvement on c_4 is very small.

3) Comparison on Beijing Dataset: As shown in Table VII and Fig. 11, our model obtains the best classification results in many categories. They are c_3:Bareland, c_4:Low_Vegetation, c_6:High_Vegetation, c_8:Dense Low-building, c_{10}:White Building. c_3 is usually widely distributed, c_4 and c_5 have blurred boundaries, and c_8 and c_{10} are densely distributed. They belong to the hard-to-classify category, but I^3RL-Net performs ingenious multisource interaction in the feature extraction process while effectively suppressing information redundancy.

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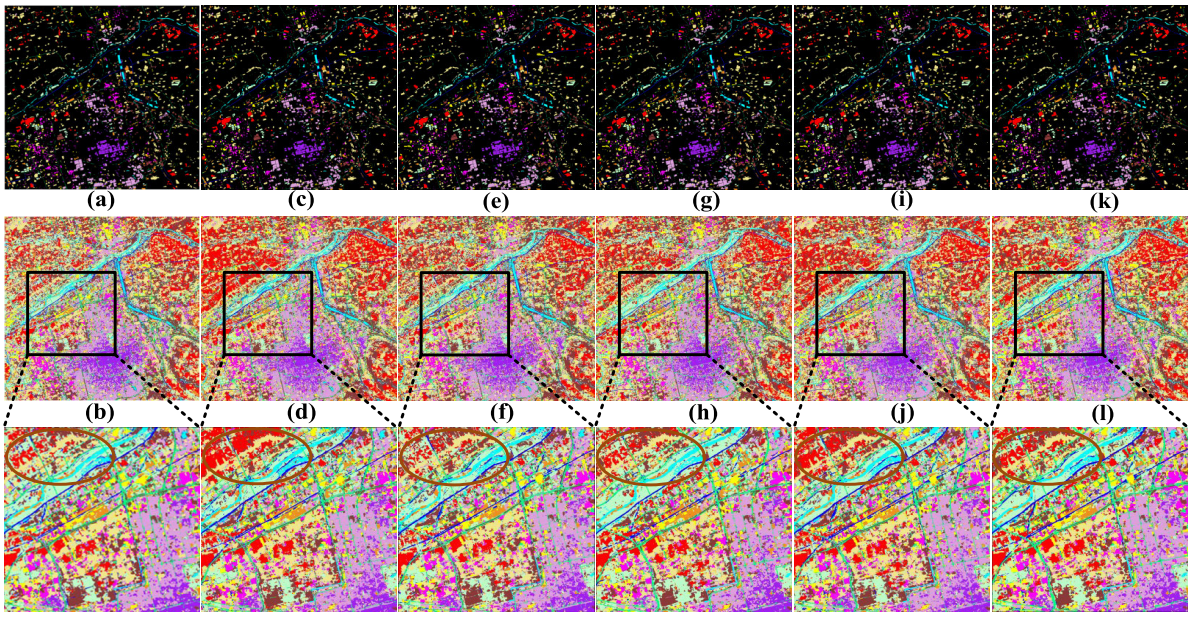


Fig. 10. Classification maps of different models on Xi'an dataset. The left column is GT, and the right column is the overall image. (a) and (b) AHF-Net. (c) and (d) SCPF-ResNet. (e) and (f) MPCA-Net. (g) and (h) CRHFF. (i) and (j) STGC-Net. (k) and (l) I^3RL-Net.

TABLE VI

QUANTITATIVE RESULTS ON XI'AN DATASET

Classes	AHF-Net	SCPF-ResNet	MPCA-Net	CRHFF	STGC-Net	OURS
c_1	99.02±0.12	99.57±0.09	99.39±0.24	99.63±0.13	99.83±0.16	**99.92±0.12**
c_2	98.54±0.32	98.41±0.28	97.46±0.12	96.85±0.16	98.27±0.20	**98.97±0.13**
c_3	93.12±0.14	97.08±0.12	96.09±0.23	97.23±0.21	97.83±0.21	**99.21±0.15**
c_4	96.07±0.09	94.92±0.15	93.25±0.06	96.58±0.07	**97.05±0.08**	96.13±0.26
c_5	94.49±0.16	93.93±0.23	92.59±0.16	93.14±0.32	96.18±0.12	**98.46±0.17**
c_6	99.09±0.24	99.08±0.21	98.40±0.11	98.38±0.12	99.13±0.28	**99.76±0.14**
c_7	95.93±0.27	**98.91±0.12**	97.95±0.14	96.30±0.11	97.78±0.31	98.58±0.14
c_8	97.41±0.21	94.52±0.15	95.93±0.32	97.50±0.24	95.62±0.11	**98.91±0.19**
c_9	97.89±0.17	96.85±0.17	98.97±0.21	97.61±0.26	97.22±0.25	**99.49±0.11**
c_{10}	97.96±0.19	99.18±0.12	98.58±0.19	97.50±0.21	99.38±0.27	**99.76±0.08**
c_{11}	98.14±0.23	99.04±0.23	99.02±0.23	98.62±0.11	99.31±0.10	**99.71±0.10**
c_{12}	96.38±0.10	95.85±0.25	98.84±0.10	95.55±0.19	98.20±0.17	**99.50±0.25**
OA(%)	96.59±0.14	97.55±0.10	97.20±0.16	97.23±0.17	98.18±0.14	**99.25±0.13**
AA(%)	96.18±0.17	97.27±0.15	97.34±0.17	97.07±0.15	97.98±0.16	**99.03±0.11**
Kappa(%)	95.91±0.15	97.23±0.16	97.08±0.14	96.86±0.18	97.94±0.17	**99.15±0.14**
Time(s)	1252.16±60.77	507.55±47.88	1321.24±75.42	601.84±36.57	**245.41±20.14**	1266.21±66.58

TABLE VII

QUANTITATIVE RESULTS ON BEIJING DATASET

Classes	AHF-Net	SCPF-ResNet	MPCA-Net	CRHFF	STGC-Net	OURS
c_1	88.10±0.27	84.07±0.13	91.57±0.26	89.90±0.16	**94.34±0.24**	89.06±0.25
c_2	80.17±0.25	80.79±0.15	**92.71±0.21**	88.34±0.23	82.08±0.13	87.46±0.20
c_3	82.17±0.14	87.20±0.19	90.43±0.15	91.20±0.12	92.25±0.19	**93.57±0.11**
c_4	87.43±0.19	78.05±0.08	89.76±0.19	88.97±0.21	85.99±0.07	**95.17±0.13**
c_5	66.02±0.38	59.13±0.25	86.29±0.10	**87.53±0.27**	70.79±0.17	87.48±0.26
c_6	91.13±0.12	90.21±0.27	90.57±0.27	90.01±0.29	91.80±0.11	**95.94±0.12**
c_7	85.29±0.19	84.98±0.31	90.46±0.12	93.28±0.24	**96.18±0.25**	95.72±0.24
c_8	85.23±0.22	89.40±0.13	92.48±0.25	94.79±0.12	94.28±0.18	**96.84±0.17**
c_9	80.63±0.24	69.34±0.11	89.96±0.29	90.06±0.08	**90.64±0.14**	86.39±0.28
c_{10}	79.67±0.13	53.14±0.18	87.27±0.14	86.63±0.22	80.60±0.29	**95.80±0.13**
OA(%)	85.95±0.20	86.24±0.11	90.26±0.17	90.13±0.19	93.67±0.15	**95.74±0.11**
AA(%)	76.54±0.18	77.63±0.13	90.15±0.14	90.07±0.22	90.01±0.18	**92.34±0.10**
Kappa(%)	78.65±0.19	79.08±0.14	89.72±0.16	90.09±0.19	90.52±0.17	**93.59±0.12**
Time(s)	3002.40±258.32	4328.16±270.84	2049.36±222.63	4096.74±297.34	**1876.38±217.65**	3747.21±256.73
Params(n)	3.94×10^6	4.27×10^6	3.98×10^6	3.41×10^6	1.68×10^7	4.76×10^6

These strategies emphasize source-specific features and better distinguish these categories.

STGC-Net achieves the highest results on c_1:Water, c_7:Dense High-building, c_9:Red Building. The distribution of

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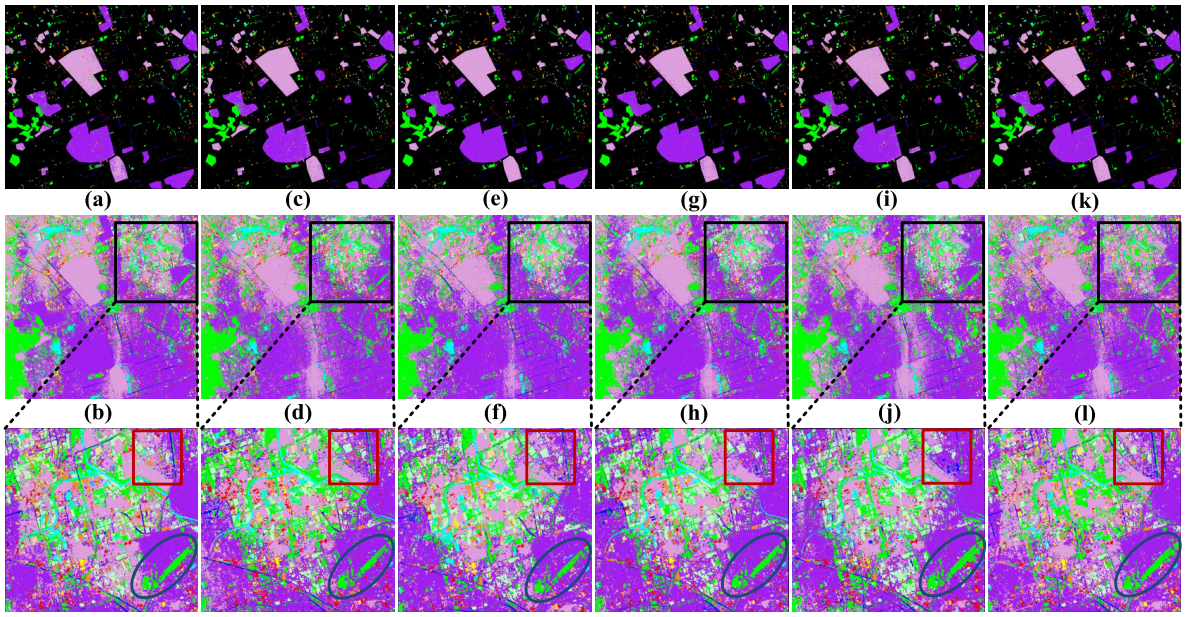


Fig. 11. Classification maps of different models on Beijing dataset. The left column is GT, and the right column is the overall image. (a) and (b) AHF-Net. (c) and (d) SCPF-ResNet. (e) and (f) MPCA-Net. (g) and (h) CRHFF. (i) and (j) STGC-Net. (k) and (l) I^3RL-Net.

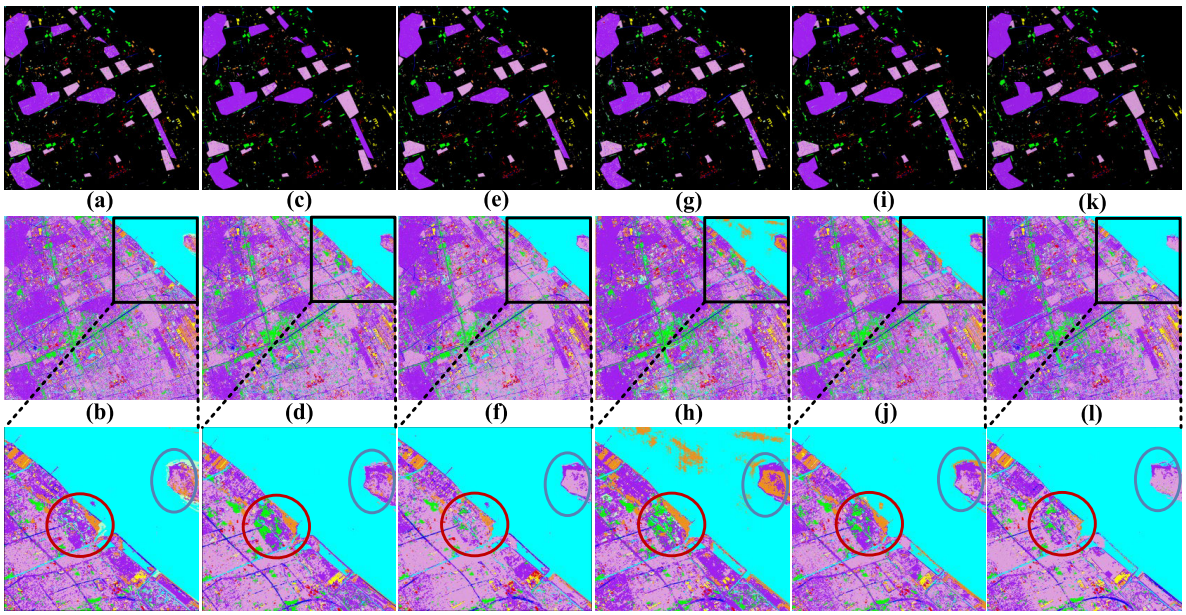


Fig. 12. Classification maps of different models on Shanghai dataset. The left column is GT, and the right column is the overall image. (a) and (b) AHF-Net. (c) and (d) SCPF-ResNet. (e) and (f) MPCA-Net. (g) and (h) CRHFF. (i) and (j) STGC-Net. (k) and (l) I^3RL-Net.

c_1 and c_9 in the Beijing dataset is less, and the distribution of c_7 is wider and more continuous. At the same time, the color information of c_1 is rich, and the spatial structure information of c_7 and c_9 is more significant. This model designs feature extraction modules for spectral and spatial information, respectively. However, it does not emphasize the existence of intersource redundant information, which may cause some categories to be misclassified into their surrounding categories. MPCA-Net achieves much higher classification accuracy than other models on c_2:Road. We can see that the c_2 of the Beijing dataset is mostly distributed adjacently, and the ADR-SS strategy of MPCA-Net can capture rich neighborhood information. In addition, the classification accuracy of CRHFF on c_5:Blue Building is not significantly different from our I^3RL-Net.

4) Comparison on Shanghai Dataset: Table VIII and Fig. 12 show the quantitative and qualitative experimental results of our methods on the Shanghai image. Our model reaches the best accuracy on c_5:Blue Building, c_6:High_Vegetation, c_7:Dense High-building, c_8:Dense Low-building, c_{10}: White Building. Among them, the spatial structure features of building categories are similar, and I^3RL-Net realizes the mutual complementation of features between PAN and MS through the implicit feature transfer of CIEF-Module. At the same time, different building categories also have their own unique features. For example, c_7:Dense High-building is

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TABLE VIII
QUANTITATIVE RESULTS ON SHANGHAI DATASET

Classes	AHF-Net	SCPF-ResNet	MPCA-Net	CRHFF	STGC-Net	OURS
c_1	83.71±0.13	86.37±0.19	92.37±0.26	89.02±0.05	**92.38±0.19**	91.84±0.27
c_2	52.90±0.07	76.55±0.17	**93.68±0.19**	89.96±0.11	91.39±0.15	90.58±0.06
c_3	77.90±0.36	**95.68±0.09**	91.28±0.23	92.57±0.24	94.50±0.06	88.09±0.16
c_4	64.76±0.22	**94.38±0.16**	88.91±0.11	89.73±0.17	83.45±0.18	92.60±0.16
c_5	80.28±0.19	88.09±0.29	89.27±0.18	91.34±0.22	92.21±0.12	**95.17±0.26**
c_6	76.67±0.25	86.24±0.21	87.42±0.20	87.57±0.15	90.91±0.27	**94.73±0.14**
c_7	76.66±0.21	91.11±0.30	90.10±0.27	91.76±0.11	94.26±0.18	**96.73±0.17**
c_8	83.01±0.28	85.14±0.15	85.49±0.26	86.96±0.20	90.25±0.22	**95.05±0.16**
c_9	65.61±0.14	77.58±0.24	**94.57±0.17**	92.98±0.15	91.82±0.14	90.99±0.19
c_{10}	93.61±0.18	94.74±0.25	95.87±0.19	95.31±0.06	96.37±0.12	**98.20±0.22**
OA(%)	78.38±0.23	89.03±0.17	89.83±0.16	90.04±0.14	92.80±0.19	**95.76±0.15**
AA(%)	75.51±0.16	87.59±0.16	90.70±0.19	90.72±0.15	90.92±0.17	**93.40±0.07**
Kappa(%)	65.56±0.13	82.09±0.19	89.69±0.15	89.97±0.09	88.85±0.15	**93.04±0.12**
Time(s)	3100.89±370.23	2967.96±359.21	2317.1±300.72	3594.32±360.81	**1516.6±174.72**	3439.34±345.71

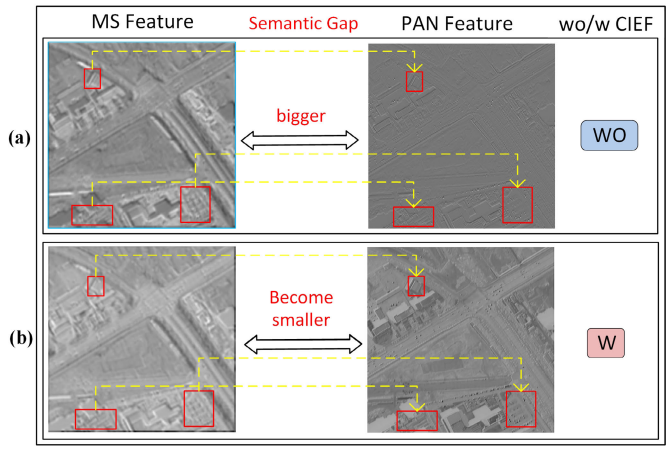


Fig. 13. Feature differences between PAN and MS patches without (wo) and with (w) CIEF-Module. (a) Semantic gap between PAN and MS features without CIEF-Module. (b) Reflects that the semantic gap between PAN and MS features is alleviated after combining CIEF-Module.

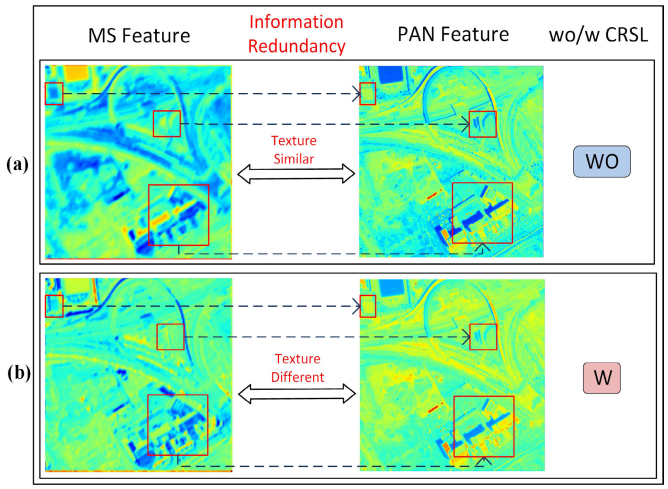


Fig. 14. Feature differences between PAN and MS patches without (wo) and with (w) CRSL-Strategy. (a) High information redundancy between PAN and MS features without CRSL-Strategy. (b) Reflects that the information redundancy between PAN and MS features is greatly reduced after combining CRSL-Strategy.

densely distributed and continuous, while the building area of c_5:Blue Building is small and scattered. CRSL-Strategy highlights the unique information of various types of buildings by reducing the information redundancy between features. The c_6:High_Vegetation distribution of MS images usually contains shadows due to illumination, while PAN images have fewer shadows. The CIEF-Module suppresses shadows in MS images through cross-source feature interaction. Thus, the highest classification accuracy is achieved on the High_Vegetation.

The classification accuracy of STGC-Net on c_1:Water is slightly higher than that of our model. c_2:Road and c_9:Red Building have complex distributions around them. The ADR-SS strategy of MPCA-Net is used to adaptively choose the suitable receptive field for feature extraction, which is conducive to dealing with complex and changing environments. Therefore, MPCA-Net achieves the best performance in these two categories. SCPF-ResNet achieves the best results on c_3:Bareland, c_4:Low_Vegetation. These two categories are usually widely distributed, but the terrain is irregular. The ASA-module of SCPF-ResNet can enhance the spatial information of objects at different scales, which is helpful in identifying the outline of objects.

G. Visualization of Feature Maps

In order to further verify the effect of our designed CRSL-Strategy and CIEF-Module, we visualize the feature changes before and after combining CRSL-Strategy and CIEF-Module.

1) Visual Validation of CIEF-Module: As shown in Fig. 13(a), PAN features are not consistent with the features expressed by MS features, especially the features marked in the red box. PAN features exhibit more contour information, while MS features emphasize texture details more. The difference in expression of shallow features will bring the semantic gap between deep features. As shown in Fig. 13(b), our CIEF-Module migrates the texture features of MS to PAN images through intersource enhanced fusion, effectively reducing the feature expression differences between shallow features and thereby alleviating the semantic gap between deep features.

2) Visual Validation of CRSL-Strategy: As shown in Fig. 14(a), the texture characteristics of PAN and MS are very similar without CRSL-Strategy, especially the features marked in the red box. It can be seen that combining CRSL-Strategy can make the features of PAN and MS become less similar. This shows that our CRSL-Strategy effectively promotes the

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complementary feature expression of PAN and MS, thereby reducing information redundancy.

V. Conclusion

This article presents an I^3RL-Net for the multisource remote sensing image classification task. After experiments, the designed network achieves satisfactory performance. CIEF-Module is designed based on the self-attention mechanism, effectively promoting the full interaction between multisource features and reducing intersource semantic differences. CRSL-Strategy is used to facilitate multisource complementary feature learning to alleviate the intersource information redundancy problem in the process of cross-source interaction. However, our model uses the self-attention mechanism in transformer in the dual branches and applies interactive modules on multilevel features, resulting in a relatively cumbersome design of the model. Shortly after, we will explore more lightweight, general, and simple structural designs.

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